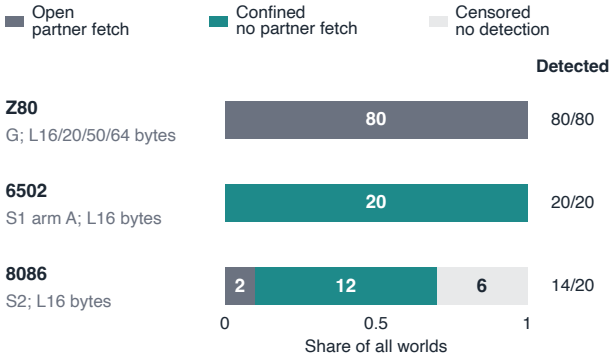
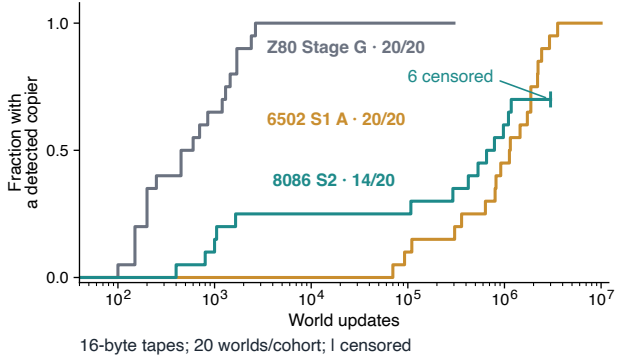
Different machines, different beginnings

a **First-replicator route**



b **Arrival: first detected heritable copier**



16-byte tapes; 20 worlds/cohort; | censored

c **Literal-channel expectation and observed beginnings**

Machine	Literal width (bytes / push)	Census: exact pusher (opcode motif)	First observed route
Z80	2	Yes · 01 C5	Open · 80/80
6502	1	No	Confined · 20/20
8086	2	Yes · B9 51	**Confined · 12/14 (2 open)**

Census candidates; first routes from evolved worlds

d **8086 convergence: the 5F A4 x y first-copier family (10/14; all ten confined)**

World seed	5F A4 = POP DI; MOVSB				x, y: variable bytes												Detection (update)	Mechanism (16/16 encounters)
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15		
8086001	5F	A4	63	3D	5F	A4	63	3D	5F	A4	63	3D	5F	A4	63	3D	652,500	One-pass copy
8086005	5F	A4	73	5D	5F	A4	73	5D	5F	A4	73	5D	5F	A4	73	5D	975,000	One-pass copy
8086006	5F	A4	6D	7D	5F	A4	6D	7D	5F	A4	6D	7D	5F	A4	6D	7D	1,650	One-pass copy
8086009	5F	A4	61	3D	5F	A4	61	3D	5F	A4	61	3D	5F	A4	61	3D	292,500	One-pass copy
8086012	5F	A4	E2	7D	5F	A4	E2	7D	5F	A4	E2	7D	5F	A4	E2	7D	1,100,000	One-pass copy
8086013	5F	A4	7D	FD	5F	A4	7D	FD	5F	A4	7D	FD	5F	A4	7D	FD	1,175,000	Copy loop
8086014	5F	A4	E9	3C	5F	A4	E9	3C	5F	A4	E9	3C	5F	A4	E9	3C	532,500	One-pass copy
8086017	5F	A4	61	BD	5F	A4	61	BD	5F	A4	61	BD	5F	A4	61	BD	422,500	One-pass copy
8086018	5F	A4	69	DD	5F	A4	69	DD	5F	A4	69	DD	5F	A4	69	DD	787,500	One-pass copy
8086020	5F	A4	63	FD	5F	A4	63	FD	5F	A4	63	FD	5F	A4	63	FD	107,500	Copy loop